

Here is the device. While B approaches A , we always *produce* AB and AD onward to fixed remote points b and d , and parallel to the secant BD we draw the line bd . Upon the distant ends we erect the auxiliary arc Acb , always kept *similar* to the shrinking arc ACB . This is the trick: the original figure dwindles toward A , but the enlarged copy out at b, d stays a fixed, finite size — a faithful magnification in which every proportion of the little figure is preserved exactly. So whatever ratio arc, chord, and tangent bear to one another below, the same ratio is held, full-sized and visible, in the finite triangle Abd and arc Acb above.